

COURSE CODE	COURSE TITLE	L	T	P	E	C
UCS2604	PRINCIPLES OF MACHINE LEARNING	3	0	0	0	3

OBJECTIVES

- To understand machine learning problems
- To study the various supervised, unsupervised and reinforcement learning algorithms in machine learning
- To study the dimensionality reduction techniques to represent the data and their dependencies.
- To understand the need of optimisation techniques.

UNIT I INTRODUCTION 8

Introduction: Machine learning; Examples of Machine Learning Applications: Learning associations – Classification – Regression – Unsupervised learning – Reinforcement learning; Preliminaries: Weight space – Curse of dimensionality – Testing machine learning algorithms – Turning data into probabilities – Basic statistics – Bias-variance tradeoff.

UNIT II SUPERVISED LEARNING 11

Neural Networks and Linear Discriminants: Brain and the neuron – Neural networks – Perceptron – Linear separability – Linear regression; Multi-layer Perceptron: Forward and Backward propagation; Support Vector Machines.

UNIT III PROBABILISTIC LEARNING, LEARNING WITH TREES 9

Probabilistic Learning: Gaussian mixture models – Nearest neighbour methods; Learning with Trees: Constructing decision trees – Classification and Regression trees – Classification example; Ensemble Learning: Boosting – Bagging – Random forests.

UNIT IV UNSUPERVISED LEARNING, REINFORCEMENT LEARNING 9

Unsupervised: K-means algorithm; Reinforcement learning: State and action space – Reward function – Discounting – Action selection – Policy – Markov decision process – Values – SARSA and Q-learning.

UNIT V DIMENSIONALITY REDUCTION, OPTIMISATION TECHNIQUES 9

Dimensionality Reduction Techniques: Linear Discriminant analysis, Principal Component Analysis; Optimisation and Search: Least-squares optimisation – Conjugate gradients – Search approaches – Exploitation and exploration.

TOTAL PERIODS: 45

COURSE OUTCOMES

On successful completion of this course, the student will be able to:

1. Explain the basic concepts of machine learning (K2)
2. Apply supervised algorithms for different classification problems (K3)
3. Explain the need for ensemble methods (K2)
4. Apply unsupervised and reinforcement learning techniques to various problems (K3)
5. Apply dimensionality reduction and optimization techniques (K3)

TEXTBOOKS

1. Stephen Marsland, “Machine Learning – An Algorithmic Perspective”, 2nd Edition, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series, 2015.
2. Ethem Alpaydin, “Introduction to Machine Learning”, 3rd Edition, The MIT Press, 2014.

REFERENCE BOOKS

1. Jason Bell, “Machine learning – Hands on for Developers and Technical Professionals”, 1st Edition, Wiley, 2014.
2. Peter Flach, “Machine Learning: The Art and Science of Algorithms that Make Sense of Data”, 1st Edition, Cambridge University Press, 2012.
3. Richert, Willi, “Building machine learning systems with Python”, Packt Publishing, 2013.
4. Tom M Mitchell, “Machine Learning”, McGraw-Hill Education (India), 2013.
5. Y S Abu-Mostafa, M Magdon-Ismail, H T Lin, “Learning from Data”, AML Book Publishers, 2012.

CO to PO Mapping

[illegible]